

Goal for Today's Meeting:

Discuss what we have done and what to do going forward.

What We Did:

- Started analysis for each aim
- Produced preliminary graphs and plots → saved in the Results/Plots directory

What We Plan to Do:

Aim 1 - Diversity Analysis

- a) Alpha Diversity
 - Plotted alpha diversity, comparing male and female within each disease group
 - Did shannon, chao1, simpson, and plan to finish the rest next week
 - Wanted to confirm if I should do adjusted p-value
 - I saw the feedback about the y-axis, but was wondering if there was any other feedback from the plots in my assignment to incorporate into the final project
- b) Beta Diversity
 - Plotted the beta diversity plot by facetting each histopathological gastric disease group and comparing each group based on biological sex
 - Performed PERMANOVA analysis for one gastric disease group as an example
 - So far, with the plot and some PERMANOVA analysis there seems to no significant difference between the sexes at each gastric disease stage
 - Once the PERMANOVA analysis was done should the p-values be adjusted?

Aim 2 - Taxonomic Differences

- a) Differential Abundance Analysis
 - Performed genus-level differential abundance analysis using ANCOM-BC2
 - Stratified samples by gastric disease stage and compared Male vs Female within each stage separately
 - Generated differential abundance results tables for each stage and visualized significant taxa using a log₂ fold-change bar plot (only the intraepithelial neoplasia stage showed significant differences, therefore was the only plot generated out of the five stages)
 - Is the current prevalence cutoff (10%) appropriate or should it be adjusted?
 - Since only one disease stage (IN) shows significant sex-associated differences, is it still biologically relevant to focus on this stage alone or should we avoid making any interpretations/conclusions about differential abundance altogether? Would it be useful to have the one plot in the final manuscript?
- b) Indicator Species Analysis
 - Performed indicator genus analysis for the patients between gastric disease stages with/without sex stratification
 - Dotplot:
 - a. Gastric disease stages (female + male)
 - b. Gastric disease stages for female and male separately
 - Which graphs will be useful to put in the final manuscript? Do we need a dotplot without sex stratification?
 - Indicator genus include "unclassified" genus → should we remove it and repeat the analysis?
 - Aside from dotplot, would boxplot for specific groups be useful to show more focused comparison?

Aim 3 - Functional Analysis

- did a literature search to find if others have found which functional pathways
- Discussed significant results with the rest of the group before the meeting

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Group 1

For Feb 27, 2026 meeting

- Should we do a heatmap or further analysis? How many pathways should be chosen so that the graph is not overcrowded?
 - Currently, they are all between 20-30 with a minimum p-adjusted value of 0.05
- Also conducted functional analysis on all stages (separated by biological sex) and compared to healthy controls
 - What should we do with these results?

Timeline + Task Assignment:

- Proposal is coming up